

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT **Paper -2(Adv-2024-P2-Model)** **Date: 21-09-2025**

Time: 02.00Pm to 05.00Pm **GTA-02** **Max. Marks: 180**

21-09-2025_Sr.Super60_NUCLEUS-BT_Jee-Adv(2024-P2)_GTA-02_Syllabus,

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type @	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Three positive real numbers x, y, z satisfy the equations $x^2+\sqrt{3}xy+y^2=25, y^2+z^2=9$ *and* $x^2+xz+z^2=16$. Then the value of $xy+2yz+\sqrt{3}xz$ is

A) 18 **B)** 24 **C)** 30 **D)** 36

2. Let A be a square matrix of order 2 with $|A|\neq 0$ such that $|A+|A|adj(A)|=0$, where adj(A) is a adjoint of matrix A. Then the value of $|A-|A|adj(A)|$ is

A) 1 **B)** 2 **C)** 3 **D)** 4

3. Let $g_1,g_2,g_3:(0,\infty)\to R$ be defined by $g_1(x)=\int_0^x\prod_{i=1}^{19}(t-i)^i\,dt, x>0$

And $g_2(x)=98(x-1)^{50}-100(x-1)^{49}+2450, x>0$

And $g_3(x)=(x-1)^{2020}(x-2)^{2021}(x-3)^{2022}, x>0$

Let α_i *and* β_i respectively denote the number of points of local minima and the number of points of local maxima of the function $g_i, i=1,2,3$ *in* $(0,\infty)$ and if the value $\alpha_1\beta_1+\alpha_2\beta_2+\alpha_3\beta_3$ *is* λ, *then* $\lambda-20$ is equal to (where $\prod_{i=1}^{n}a_i=a_1a_2a_3.....a_n, n\in N$

A) 7 **B)** 5 **C)** 8 **D)** 9

4. Let P(x) be a polynomial of degree less than 100 such that $P(1), P(-1)$ *and* $P(0)$ are distinct non-zero quantities. If $\frac{d^{100}}{dx^{100}}\left(\frac{P(x)}{x^3-x}\right)=\frac{f(x)}{g(x)}$ for some polynomials f(x) and g(x) then the value of $\frac{n}{4}$ is (where 'n' is the smallest possible degree of f(x))

A) 100.75 **B)** 52.50 **C)** 50.25 **D)** None of these

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. The value(s) of 'p' for which the equation $ax^2 - px + ab = 0$ *and* $x^2 - ax - bx + ab = 0$ may have a common root, given a, b are non zero real numbers, is:

A) $a + b^2$ **B)** $a^2 + b$ **C)** a(1 + b) **D)** b(1 + a)

6. Let a, b, c, d be distinct integers such that the equation $(x-a)(x-b)(x-c)(x-d)-9=0$ has an integer root 'r', then the value of $a+b+c+d-4r$ is less than

A) 0 **B)** 1 **C)** 2 **D)** 3

7. Let $\bar{x}, \bar{y}, \bar{z}$ be three vectors, such that $|\bar{x}| = |\bar{y}| = |\bar{z}| = \sqrt{2}$ and $\bar{x}, \bar{y}, \bar{z}$ make angles of 60° with each other also $\bar{x} \times (\bar{y} \times \bar{z}) = \bar{a}$, $\bar{y} \times (\bar{z} \times \bar{x}) = \bar{b}$ *and* $\bar{x} \times \bar{y} = \bar{c}$, then which of the following is/are true?

A) $\bar{x} = \frac{1}{2}\left[(\bar{a}+\bar{b}) \times \bar{c} - (\bar{a}+\bar{b})\right]$ **B)** $\bar{y} = \frac{1}{2}\left[(\bar{a}+\bar{b}) + (\bar{a}+\bar{b}) \times \bar{c}\right]$

C) $\bar{z} = \frac{1}{2}\left[(\bar{b}-\bar{a}) + (\bar{a}+\bar{b}) \times \bar{c}\right]$ **D)** $\bar{x} + \bar{y} + \bar{z} = (\bar{a}+\bar{b}) \times \bar{c}$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

8. It is given that $f(x)$ is defined on R satisfying $f(1) = 1$ *and for* $\forall\, x \in R$,

$f(x+5) \geq f(x) + 5$ *and* $f(x+1) \leq f(x) + 1$. *If* $g(x) = f(x) + 1 - x$, then find the value of

$g(2025)$ (where $f(x)$ is differentiable)

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9. The equation of plane of a ΔABC is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=\lambda$ and lengths of sides of ΔABC opposite to vertices A, B, C are respectively p, q, r. Let D, E, F are mid points of sides BC, CA, AB respectively and D, E, F lies on x-axis, y-axis, z-axis respectively. If $a^2=\frac{q^2+r^2-p^2}{2}, b^2=\frac{p^2+r^2-q^2}{2}$ *and* $c^2=\frac{p^2+q^2-r^2}{2}$ then the value of $\left|\frac{1}{\lambda}\right|$ is

10. Consider $f(x)=\frac{x^2+4}{|x|}$ and $g(x)=p|x|+2, p>0$. If the exhaustive set of values of 'p' for which $f(x)>g(x)\forall x\in R-\{0\}$ *is* (a,b), *then* $a+4b$ *is*

11. How many of the following statements are correct?

A) Adjoint for a 3×3 singular matrix can't be defined

B) If adj $A=\begin{bmatrix}4 & 2022 & 4\\ 2020 & 4 & 2021\\ 2023 & 4 & 2024\end{bmatrix}$; then the number of possible real matrices "A" is 2

C) If adjA= $\begin{bmatrix}2020 & 2021\\ 2023 & 2024\end{bmatrix}$ then the number of possible real matrices "A" is 1.

D) If adjA= $\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$; then the number of possible real matrices "A" is not finite

12. Let for $x\in R, S_0(x)=x,\ S_K(x)=x.C_k+k\int_0^x S_{K-1}(t)dt$, *where* $C_0=1, C_k=1-\int_0^1 S_{K-1}(x)dx, k=1,2,3....$ *Then* $S_2(3)+6C_3$ *is equal to*

13. Find the value of $\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\sum_{n=0}^{\infty}\frac{1}{k2^n+1}$

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**

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- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Let $g(x)=6x^2-18x+8;\ f_1(x)=|g(x)|, f_2(x)=|f_1(x)-P_1|, f_3(x)=|f_2(x)-P_2|, f_4(x)=|f_3(x)-P_3|.$

14. If $P_1=7$, then the largest integral value of P_2 such that $f_3(x)$ has exactly 10 points of non-differentiability, is

15. If P_1,P_2,P_3 are consecutive primes such that $5<P_1<P_2<P_3$ and total number of non-diffententiable points of f_1,f_2,f_3 are a_1,a_2,a_3 respectively, then the number of solution of equation $\dfrac{(x-a_1)(x-a_3)}{(x-a_2)}=1, is$

PARAGRAPH-II

Let us define a function $g(x)=\begin{cases}x^2 & 0\le x<1/2\\ (x-1)^2; & \frac{1}{2}\le x<1\end{cases}$ and another function f(x) such that the

Area enclosed by curves g(x) and x-axis between the ordinates $x=\beta\ and\ x=\beta+f(\beta)$ is half the area enclosed by g(x) and x-axis from $x=0\ to\ x=1$(where $0<\beta<\frac{1}{2} and \frac{1}{2}<\beta+f(\beta)<1$).

16. The value of $\underset{x\to\frac{1}{2}}{Lim}\dfrac{(f(x)+x-1)^3}{\left(x-\frac{1}{2}\right)}$ is L then find the value of 4L

17. Let at $x=\alpha, f(x)$ has local extremum, then the value of $32\alpha^3$ is

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PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

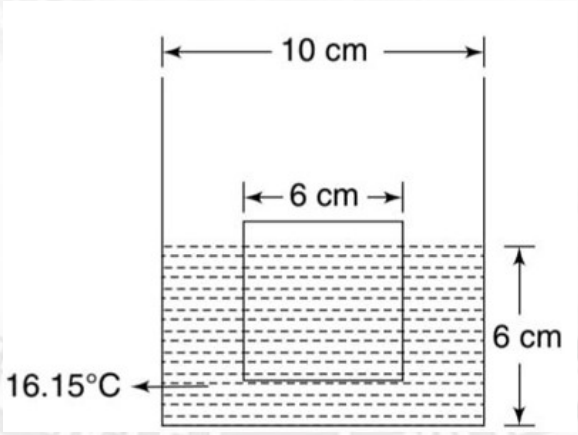
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Negative Marks : −1 In all other cases

18. A container has a square cross-section of 10 cm $\times$ 10 cm. A cubical ice block of side length 6 cm is floating in water in the container. Water level in the container is 6 cm high. The ice block is at a temperature of 0^0 C and the water is at 16.15^0 C Assume that heat exchange take place between the ice block and water only. What length of ice block will remain submerged in water when the system reaches thermal equilibrium? Assume that the ice block maintains its cubical shape as it melts. Take density of ice = 0.9g /cc, density of water = 1.0 g /cc

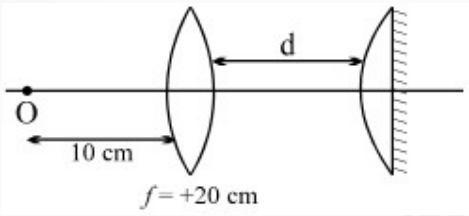
Specific heat capacity of water $= 1 cal\ g^{-1}\ {}^{o}C^{-1}$, Specific latent heat of fusion of ice $= 80 cal\ g^{-1}$



A) 4.5 cm **B)** 5.5 cm **C)** 3 cm **D)** 5 cm

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19. A convex lens of focal length 20 cm and another Plano-convex lens of focal length 40 cm are placed co-axially (see figure). The Plano-convex lens is silvered on plane surface. What should be the distance d (in cm) so that final image of the object 'O' is formed on O itself ?



A) 10 **B)** 15 **C)** 20 **D)** 25

20. Due to a vertical temperature gradient in the atmosphere the index of refraction varies. Suppose index of refraction varies as $n = n_0\sqrt{1+ay}$ where n_0 is the index of refraction at the surface and $a = 2.0\times10^{-6}\,m^{-1}$. A person of height $h = 2m$ stands on a level surface. Beyond what distance he cannot see the runway?

A) 100m **B)** 2000m **C)** 1000m **D)** 200m

21. The length of elastic string, obeying Hooke's law is l_1 metres when the tension 4 N and l_2 metres when the tension is 5 N. The length in metres when the tension is 9 N is:

A) $5l_1 - 4l_2$ **B)** $9l_1 - 8l_2$ **C)** $5l_2 - 4l_1$ **D)** $9l_2 - 8l_1$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

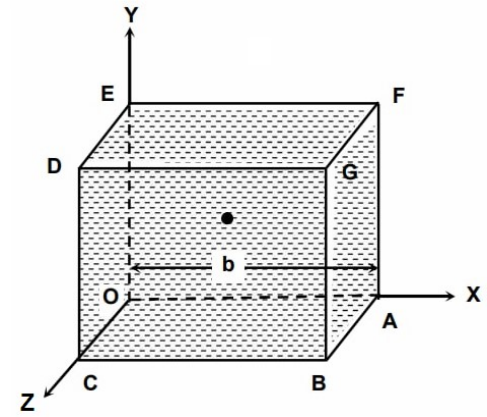
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22. A liquid of density ' ρ ' is filled in a cubical container OABCDEFG. The point 'O' is assumed to be origin. A small ball of density ρ_0 is kept at $\left(\frac{b}{2},\frac{b}{2},\frac{b}{2}\right)$ as shown in figure and an acceleration $2g\hat{i}+g\hat{j}$ is given to container. Uniform gravity exists along negative y- axis. Now, choose the correct option(s).



A) If $\rho=\rho_0$ then acceleration of small ball is $\sqrt{5}g$.

B) If $\rho=2\rho_0$ then small ball will hit the mid of edge GF.

C) If $\rho=3\rho_0$ then small ball will hit the mid of edge GF.

D) If $\rho=\rho_0$ then net force on the small ball is zero.

23. A string of mass m is fixed at both its ends. The fundamental mode of string is excited and it has an angular frequency ω and the maximum displacement amplitude A. Then :

A) The maximum kinetic energy of the string is $E_K=\frac{1}{4}mA^2\omega^2$

B) The maximum kinetic energy of the string is $E_K=\frac{1}{2}mA^2\omega^2$

C) The mean kinetic energy of the string averaged over one periodic time is

$<E_K>=\frac{1}{4}mA^2\omega^2$

D) The mean kinetic energy of the string averaged over one periodic time is

$<E_K>=\frac{1}{8}mA^2\omega^2$

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24. Imagine a Young's double slit interference experiment performed with electron waves associated with fast moving electrons coming from the electron gun. The distance between successive maximum will decrease if the:

A) Accelerating voltage in electron gun is decreased

B) Accelerating voltage is increased

C) Distance between screen and slits is increased

D) Distance between the slits is increased

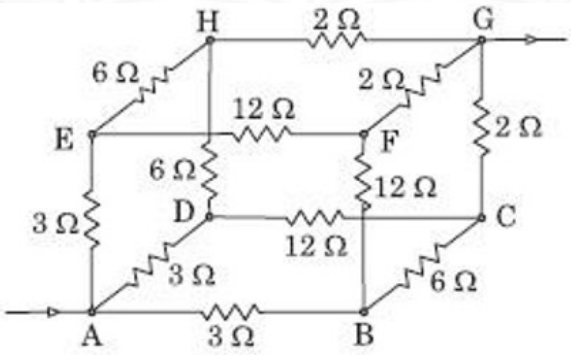
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
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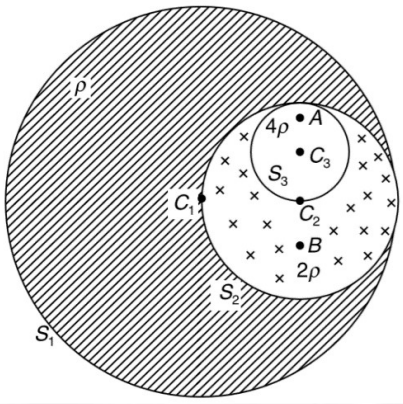
25. An astronaut on the surface of the moon throws a piece of lunar rock (mass m) directly towards the earth at a great speed such that the rock reaches the earth. Mass of the earth = M, Mass of the moon $=\frac{M}{81}$ Radius of the earth = R, Distance between the centre of the earth and the moon = 60R.In the course of its journey the maximum gravitational potential energy of the rock is given by $-\frac{nGMm}{243R}$.The value of n

26. Twelve resistances are arranged in the edges of a cube and electrically connected only at the corners of the cube. Find equivalent resistance between the corners A and G (in Ω).[Mark the nearest integer]

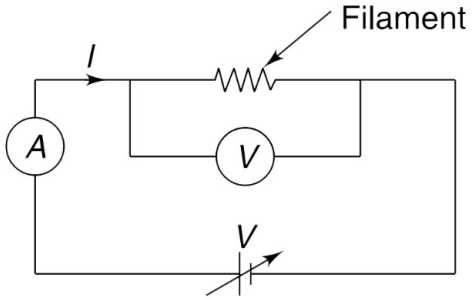


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27. In the figure shown, spheres S_1, S_2 *and* S_3 have radii $R, \frac{R}{2}$ *and* $\frac{R}{4}$ respectively. C_1, C_2 *and* C_3 are centers of the three spheres lying in a plane. Angle $\angle C_1C_2C_3$ is right angle. Sphere S_3 has a uniformly spread volume charge density 4ρ. The remaining part of S_2 has uniform charge density of 2ρ and the left over part of S_1 has a uniform charge density of ρ. The electric field at a point A at a distance $\frac{R}{8}$ from C_3 on the line C_2C_3 is given by $\frac{\sqrt{x}}{6}\frac{\rho R}{\epsilon_0}$ the value of x

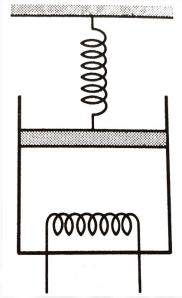


28. A tungsten filament bulb is connected to a variable voltage supply. The potential difference, V is varied and the current, I and steady temperature T of the filament is recorded. A graph is plotted for ln (VI) Vs ln (T). Find the slope of the graph. Assume that temperature of filament $T >> T_0 =$ atmospheric temperature.



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29. 2.00 mol of a monatomic ideal gas (U = 1.5 nRT) is enclosed in an adiabatic, fixed, vertical cylinder fitted with a smooth, light adiabatic piston. The piston is connected to a vertical spring of spring constant $200 Nm^{-1}$ as shown in figure. The area of cross section of the cylinder is 20.0 cm². Initially, the spring is at its natural length and the temperature of the gas is 300 K. The atmospheric pressure is 100 kPa. The gas is heated slowly for some time by means of an electric heater so as to move the position up through 10cm. Find the work done by the gas (in J)



30. Two cylinders A and B fitted with pistons contain equal amounts of an ideal diatomic gas at 300 K. The piston of A is free to move while that of B is held fixed. The same amount of heat is given to the gas in each cylinder. If the rise in temperature of the gas in A is 30 K, then calculate the rise in temperature of the gas in B in kelvin.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

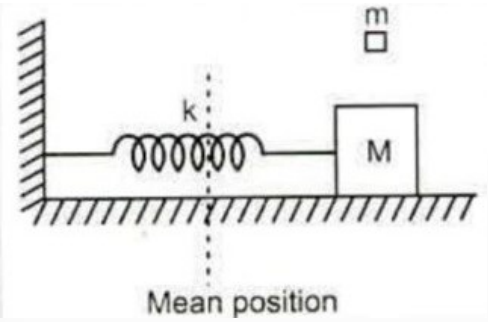
A horizontal spring block system executes S.H.M. with amplitude A = 10 cm, initial phase $\phi = 0$ and angular frequency ω. The mass of block is M = 13kg and there is no friction between the block and the horizontal surface. The spring constant being 2500 N/m.

At $t = t_1$ sec [for which $\omega t_1 = \phi_1 = 30^o$] . A mass m = 12kg is gently put on the block.

[Assume that collision between the block and the mass is perfectly inelastic and mass m remains stationary w.r.t. the block M always].

Read above passage carefully and answer the following questions:

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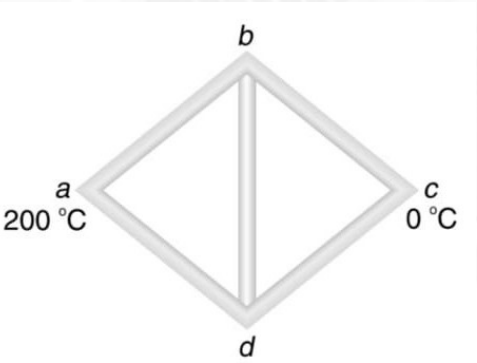


31. The new angular frequency of the system will be (in rad/sec)

32. The new amplitude of system (in cm)

PARAGRAPH-II

Five rods of same material and same cross section are joined as shown. Lengths of rods ab, ad and bc are $1m, 2m$ and $3m$ respectively. Ends a and c are maintained at temperatures $200^0\ C$ and $0^0 C$ respectively.



Based on the above facts, answer the following questions.

33. The length of rod dc for which there will be no heat flow through rod bd is (in m)

34. If there is no heat flowing through rod bd then temperature of junction b or d is (in $^\circ C$)

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CHEMISTRY **Max Marks: 60**

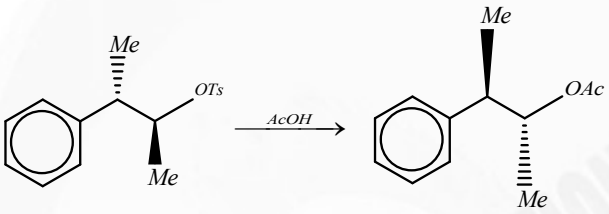
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

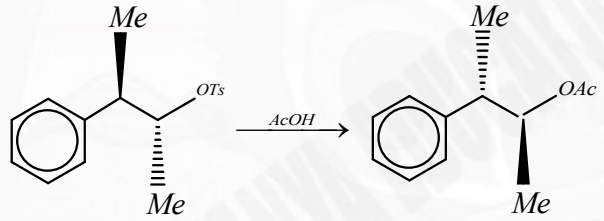
This section contains **FOUR (04)** questions.

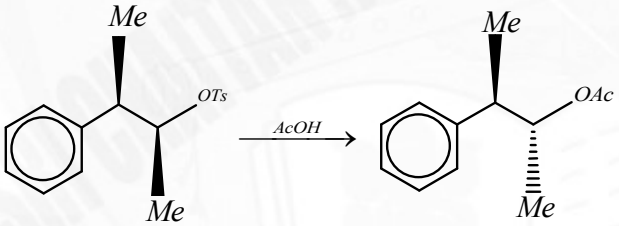
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

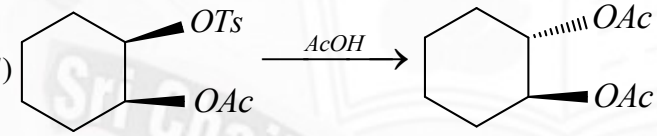
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. In how many of the following reactions, product (major) is(are) wrongly given ?

P) 

Q) 

R) 

S) 

A) 1 **B) 2** **C) 3** **D) 4**

36. Compute $\Delta_r G$ in KJ/mole (approx) for the reaction

$$H_2O(l,\ 1\ atm,\ 323\ K) \rightarrow H_2O\ (g,\ 1\ atm,\ 323\ K)$$

Given that: $\Delta_{vap}H\ at\ 373\ K = 40\ kJmol^{-1}$, $C_p\left(H_2O, l\right) = 75\ JK^{-1}mol^{-1}$,

$C_p\left(H_2O,\ g\right) = 35\ JK^{-1}mol^{-1}$. $\left[Given\ ln\left(\frac{323}{373}\right) = -0.14\right]$

A) 3.6 **B)** 5.6 **C)** 7.6 **D)** 9.6

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37. A solution was prepared by dissolving 2g of a non-volatile non electrolyte solid **B** in 100g of solvent **A**. At 354K, the vapour pressure of the solution became exactly equal to the atmospheric pressure of 1.0 atm. If the molar mass of **A** is 80 g mol^{-1}, the molar mass (in g mol^{-1}) of solid **B** is approximately

[Given: Gas constant, $R = 8.3 JK^{-1}mol^{-1}$; Vapour pressure of A at 353K = 1.0 atm, Standard entropy of vapourisation of A at $353K = 85 JK^{-1}mol^{-1}$,]

A) 55.25 **B)** 58.25 **C)** 61.25 **D)** 64.25

38. When N_2O_5 is heated at temp. T, it dissociates as $N_2O_5 \rightleftharpoons N_2O_3 + O_2$, $K_c = 2.5$. At the same time N_2O_3 also decomposes as : $N_2O_3 \rightleftharpoons N_2O + O_2$.. If initially 4.0 moles of N_2O_5 are taken in 1.0 litre flask and allowed to attain equilibrium, concentration of O_2 was formed to be 2.5 M. Equilibrium concentration of N_2O is

A) 1.0 **B)** 1.5 **C)** 2.166 **D)** 0.334

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. On alternating polymerisation of "x" unit of Ethenyl Ethanoate with Ethene a polymer "A" is obtained.

$$A \xrightarrow{Hydrolysis} B \xrightarrow[Excess]{CrO_3} C \xrightarrow[(ii)H_2O]{(i)CH_3-MgBr(Excess)} D \xrightarrow[(Excess)]{Conc.H_2SO_4}$$

$$E \xrightarrow[Zn,H_2O]{O_3} F \xrightarrow[\Delta]{NH_3.H^+} G \xrightarrow{CHCl_3/NaOH} H + I$$

If "H" gives ppt of AgCl in carius method, Then correct statement is/are.

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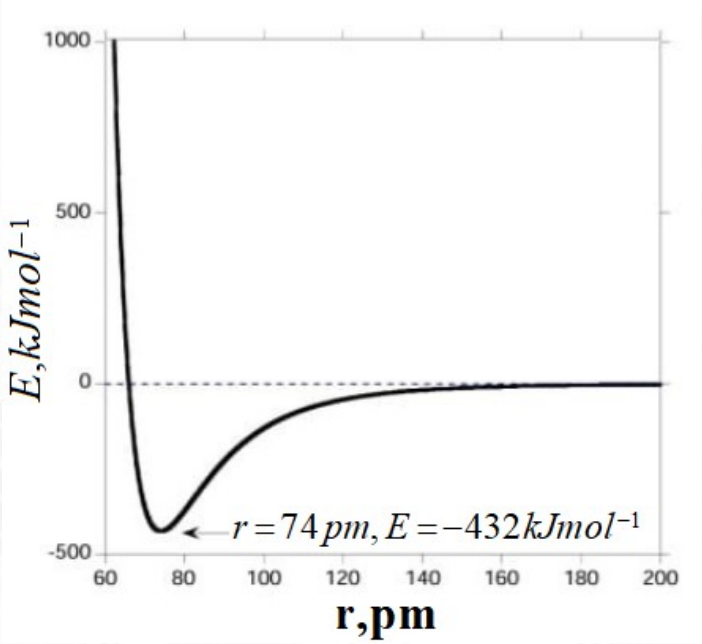
A) $I > G > H$ (Basic strength order)

B) E may be natural Rubber

C) B can be obtained by polymerisation of Vinyl alcohol & ethene

D) If B on reaction with excess of CH_3COCl increases its mass by 21000 then 500 units of ethene is used in preparation of "A".

40. The figure given below is the plot of potential energy E of two hydrogen atoms as a function of their internuclear separation r. Choose the correct statement(s)



A) The potential energy increases as the internuclear separation decreases from 70 pm to 60 pm due to the electron-electron repulsion which overwhelms the electron nucleus attraction

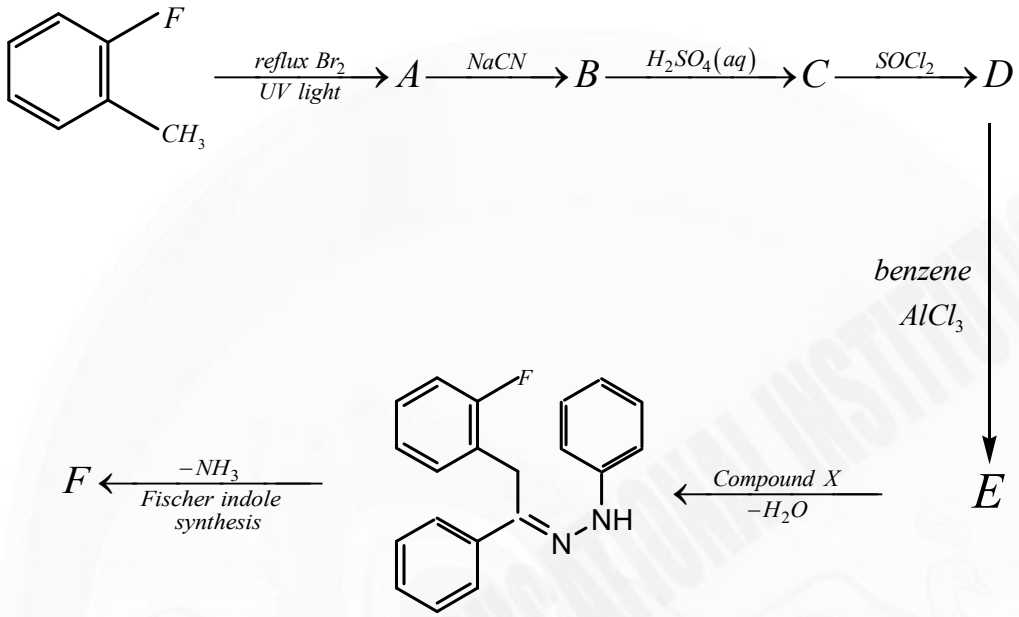
B) The potential energy decreases as the internuclear separation decreases from 120 pm to 80 pm due to the electron-nucleus attraction between the two atoms

C) The longest wavelength of light that has sufficient energy to break a H-H bond equals 312 nm

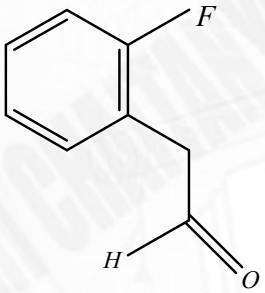
D) The bond dissociation enthalpy of the H-H bond is 216 kJ mol^{-1}

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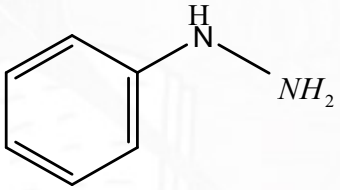
41. Consider the reaction scheme shown below where A – F are the major products.



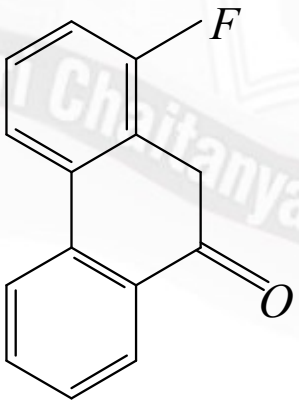
Choose the correct options(s)



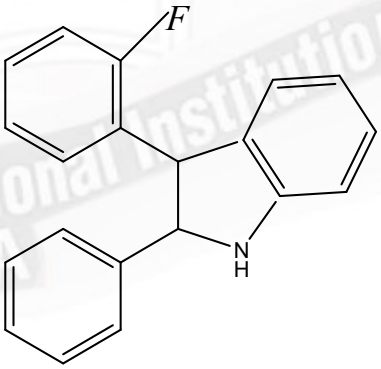
A) **C** is



B) **X** is



C) **E** is



D) **F** is

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SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

42. $[Fe(H_2O)_6]^{3+}, [Mn(H_2O)_6]^{3+}, [Co(H_2O)_6]^{3+}, [Co(NH_3)_6]^{2+}, [Co(NH_3)_6]^{3+}, [Co(NO_2)_6]^{3-},$

$[NiF_6]^{2-}, [Mn(CN)_6]^{3-}, [Fe(C_2O_4)_3]^{3-}, [Fe(acac)_3], [Fe(bpy)_3]^{3+}, [Fe(phen)_3]^{3+}, [Fe(en)_3]^{3+}, [Fe(en)_3]^{2+},$

$[Fe(phen)_3]^{2+}, [Fe(bpy)_3]^{2+}, [CoF_6]^{2-}, [CoF_3(H_2O)_3], [Cr(CN)_6]^{4-}, [RuCl_6]^{2-}, [Co(NH_3)_4(NO_3)_2]^{+}$

How many of the following is/are low spin ?

43. Choose the number of correct statement(s) from the following ?

a) Order of ionization energy: $N_2 > N > O > O_2$

b) Highest occupied molecule orbital (HOMO) of CO molecule is non bonding MO with slight antibonding character.

c) Correct solubility in water order: $Na_2CO_3 < K_2CO_3 < Li_2CO_3$

d) Thermal stability order : $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$

e) Correct order of bond angle : $NO_2^+ > NO_2^- > NO_2$

f) Lewis acidity : $BCl_3 > AlCl_3$

g) Lewis basicity: $N(CH_3)_3 > N(SiH_3)_3$

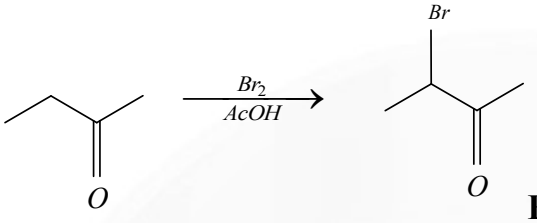
h) Dipole moment as well as bond angle: $NH_3 > NF_3$

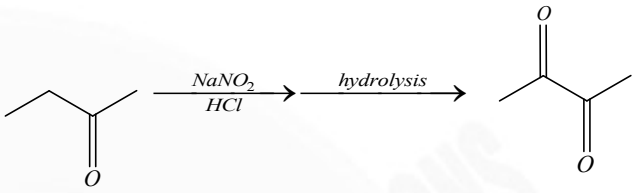
i) pK_a order: $CHF_3 < CHCl_3$

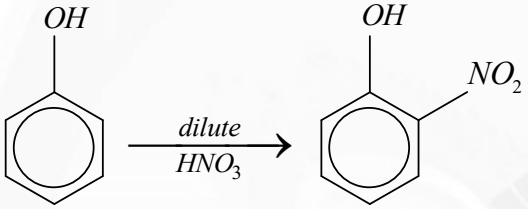
j) CO bond order: $CO > H_3B \leftarrow CO > (Mn(CO)_5)^-$

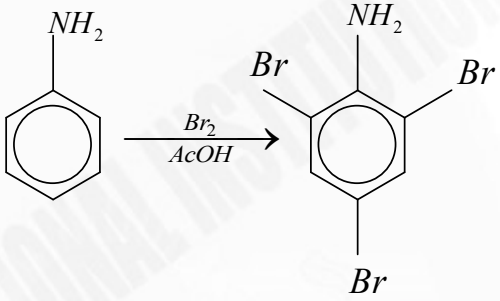
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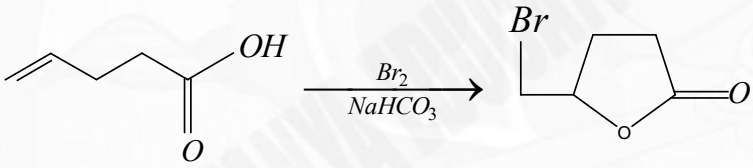
44. Choose the number of correct statement(s) from the following related to major product ?

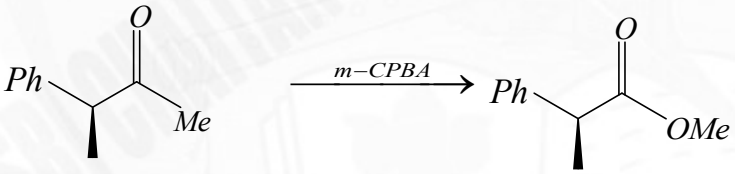
A) 

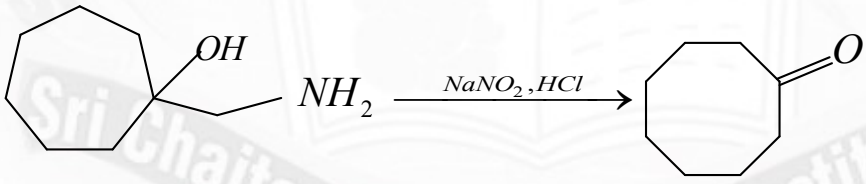
B) 

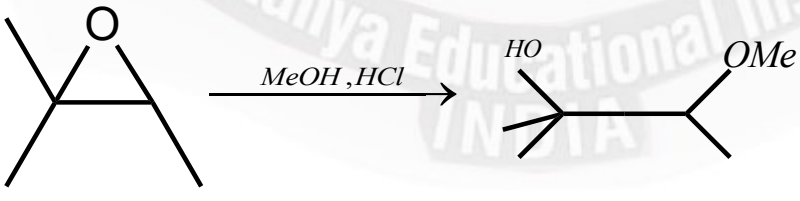
C) 

D) 

E) 

F) 

G) 

H) 

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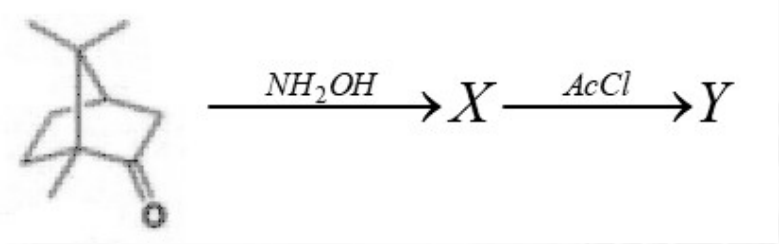
45.



Degree of unsaturation of A=x

Oxygen atoms present in A=y, $x + y =$?

46.



Consider major in each case.

Degree of unsaturation in Y=a

Number of sp^2 carbons in Y = b. Find a + b

47. **X, Y** and **Z** are atoms of three non metallic elements belonging to period 2 of the modern periodic table. The relative variation in the covalent radius (r_{cov}), the first ionization enthalpy $(\Delta_i H_1)$ and the electron affinity (A_e) of **X, Y** and **Z** are displayed below. If the atomic number of **X, Y** and **Z** are $n-1$, n *and* $n+1$, respectively, then the value of n is_____

r_{cov} : $Z < Y < X$

$\Delta_i H_1$: $Z > X > Y$

A_e : $X < Y < Z$

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

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PARAGRAPH-I

Consider a galvanic cell constructed at 298 K with one half-cell consisting of a 10 g silver wire immersed in 1 L of a 0.10 M solution of silver nitrate. The second half-cell consists of a 20 g copper plate that is immersed in 1 L of a 0.20 M solution of copper(II) sulfate. Let the voltage measured for this galvanic cell be E Volts. If the cell is discharged at a constant current of 0.15 A, let the time taken for the mass of the silver electrode to be equal to the mass of the copper electrode be t seconds. Given $\begin{matrix} At.\,Wt\ of\ \ Ag = 108 \\ At.\,Wt\ of\ \ Cu = 63.5 \end{matrix}$

Half-reaction	E^O, V
$Ag^+(aq) + e^- \rightarrow Ag(s)$	0.800
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$	0.337

48. The value of E is _____

49. The value of $\frac{t}{1000}$ is _____

PARAGRAPH-II

24 mL of a solution containing HCl was treated with excess of 0.004M KIO_3 and KI solution of unknown concentration where I_2 liberated is titrated against a standard solution of 0.02M $Na_2S_2O_3$ solution whose 24 mL were used up..

50. Find the molarity of HCl_____

51. Find the volume(in ml) of KIO_3 solution consumed_____

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